

KongDoc - Diary App for Children with ASD

How can children with ASD receive accurate and continuous care over time?

MY ROLE

UX Designer
& Project Manager

THE TEAM

2 Medical Doctors,
2 Software Engineers

AWARDS

2019 - Dreamshare Medical Hackathon
2020 - Yonsei Start-up Incubation Program

2020 - Digital Health Hackathon (1st Place)

2021 - Kakao Impact Incubation Program

OVERVIEW

I worked on interaction design of Ballie, building trust and emotional connections with users.

At the Healthcare Entrepreneurship Society "Medilux," multi-background members interested in children with developmental disabilities gathered. Through item development, our team won the grand prize at the Digital Healthcare Hackathon and received project development investment.

BACKGROUND

My interest began through working with children with ASD.

Through volunteering with diverse groups of children, I developed a deep interest in early childhood development. By working with both typically developing children and children with autism spectrum disorder (ASD), I became increasingly aware of the challenges their parents face, particularly in accessing timely and effective support.

Volunteering with children

High Treatment Costs & Long Wait Times

The cost of effective autism treatment is extremely high. Applied Behavior Analysis (ABA), one of the most widely recognized treatment methods, can cost up to ₩4 million per month for full-time sessions. (Autism Partnership Korea)

Rising Prevalence of ASD

According to the CDC, the prevalence of autism spectrum disorder in the United States has risen sharply, from 1 in 166 (0.67%) in 2004 to 1 in 68 (1.46%) in 2016, and further to 1 in 59 (1.69%) in 2018.

Importance of Early Intervention

According to the CDC, the prevalence of autism spectrum disorder in the United States has risen sharply, from 1 in 166 (0.67%) in 2004 to 1 in 68 (1.46%) in 2016, and further to 1 in 59 (1.69%) in 2018.

PROBLEM

Fragmented behavior records disrupt clinical interpretation and continuity of care.

Problem diagram

Among the various issues related to developmental disabilities, we focused on behavioral records of children with developmental disabilities. For children with developmental disabilities, everyday behavioral observations often play an important role in clinical assessment and ongoing care. Caregivers often document behaviors through scattered photos, videos, and notes, making it difficult to identify patterns and communicate them efficiently during clinical visits.

TARGET USER GROUP

Caregivers during the initiation and management phase of developmental therapy

Target user group stages

Caregivers' needs and decision-making priorities shift depending on their psychological state, parenting stage, and position within the treatment journey.

RESEARCH

Interviews revealed that parents were too busy, records were fragmented, and they didn't know what kind of data should be shared.

Desk Research : Surveys, Developmental Disability Blogs

Field Research : Qualitative research (interview - user/expert interview)

Data Modeling : Affinity Mapping, Personas

Service Concept : Brainstorming

Service Define : user scenario, storyboard, IA

Stakeholder Definition

Rather than treating caregivers as isolated users, we mapped the full ecosystem surrounding a child's care to understand where behavioral data breaks down across institutions, roles, and responsibilities

Stakeholder ecosystem diagram

In-Depth Interviews

To understand the needs of caregivers more deeply, we conducted 1:1 interviews with parents of children with ASD.

Participants: 6 parents of children with developmental disabilities
Age of children: 3–5

Professional Research

Method: One-on-one in-depth interviews, 35 minutes per session
To understand the needs of caregivers more deeply, we conducted 1:1 interviews with parents of children with ASD. These interviews helped us see caregiving as an integrated process between emotional labor and time pressures.

Interviews: 9 interviews
Participants: 4 parents, 2 therapists (2 speech-therapists, 2 occupational therapists)

KEY INSIGHTS

Caregivers need an easier way to capture, share, and communicate a child's behavioral progress.

Guardians' parenting styles and emotional states affect their records

"When a behavioral issue happens, I'm too busy handling the situation, removing dangerous objects or calming them down..."
— Caregiver C

Multiple caregivers often share roles, leading to data fragmentation.

"My child is sometimes cared for by their grandmother or a teacher, and in those cases, it's hard to manage all the records consistently."
— Caregiver A

Uncertainty about what to report during short doctor visits

"The consultation is so short, and I never know what I should show the doctor. I just freeze up."
— Caregiver B

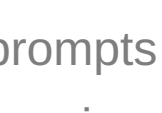
Data for both the doctor and the parents is needed. Track data for long-term treatment

Experts emphasized the importance of offering the service in a digital format that enables doctors and therapists to easily review behavior logs and track progress.

Emotional Sensitivity of Caregivers

Caregivers often experience significant emotional stress. The product should be designed to support their emotional well-being as well as provide practical tools.

Speaking with clinicians and parents revealed a critical gap: while professionals value longitudinal behavioral patterns, the data they receive from caregivers is inconsistent and fragmented, leading to data fragmentation.



Guided logging for busy routines

Reduce documentation burden through quick, structured prompts that caregivers can complete during busy routines.



Shared behavior logs and schedule management

Enable multiple caregivers to collaboratively manage a shared schedule and behavioral records. This ensures all stakeholders are "on the same page" regarding the child's developmental status.



Visualized reports for clinical communication

Provide summarized, visual reports highlighting only the most relevant data for clinical use. This helps caregivers quickly and clearly communicate during short medical appointments.

UX WRITING

I defined writing principles to provide emotional warmth and reduce the burden of documentation.

Warm encouragement for emotionally burdened users

Parents of children with developmental disabilities experience the highest levels of stress among caregivers of children with illnesses or disabilities. I recognized that warm, encouraging communication is a crucial element in supporting consistent and sustainable record-keeping.

"You're already a wonderful mom, just by being present."

"Well done! You've finished all your logs for the week."

Ask questions like a kind, smart doctor

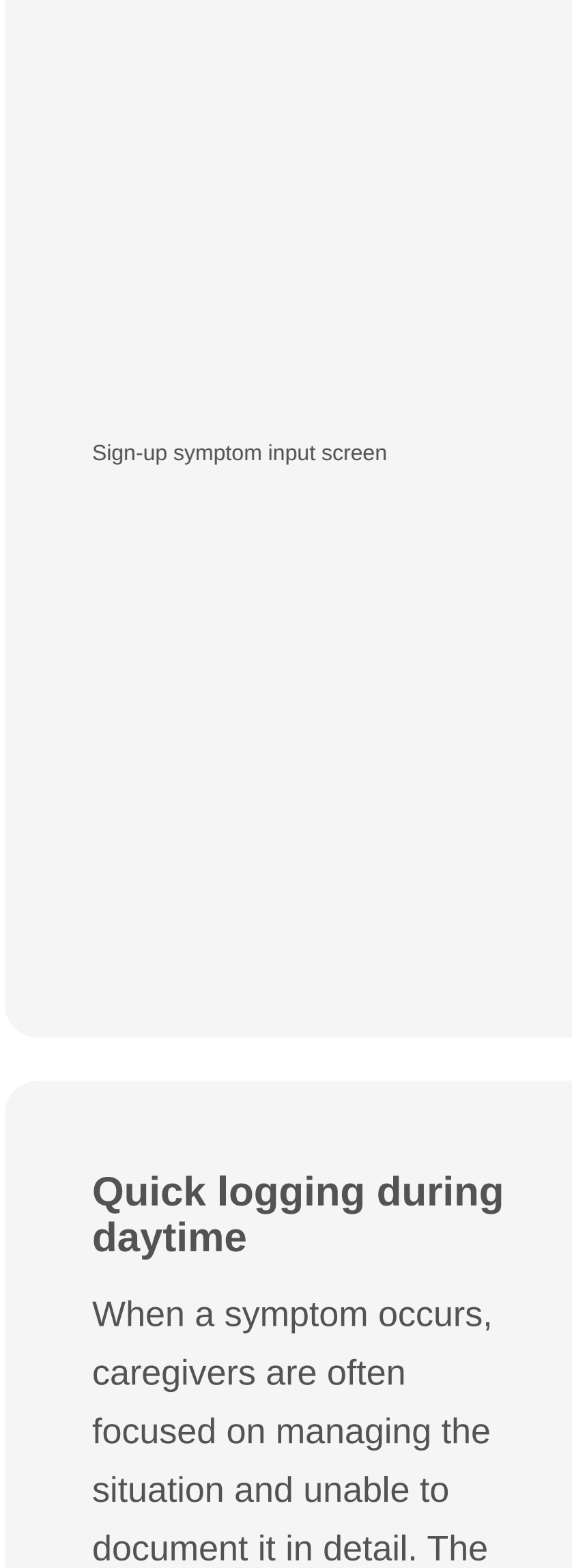
We designed our prompts to be simple and intentional—like a friendly doctor asking just the right questions—so that caregivers could feel guided, not overwhelmed.

"How did those responses from others make you feel?"

"Do these symptoms or behaviors occur repeatedly?"

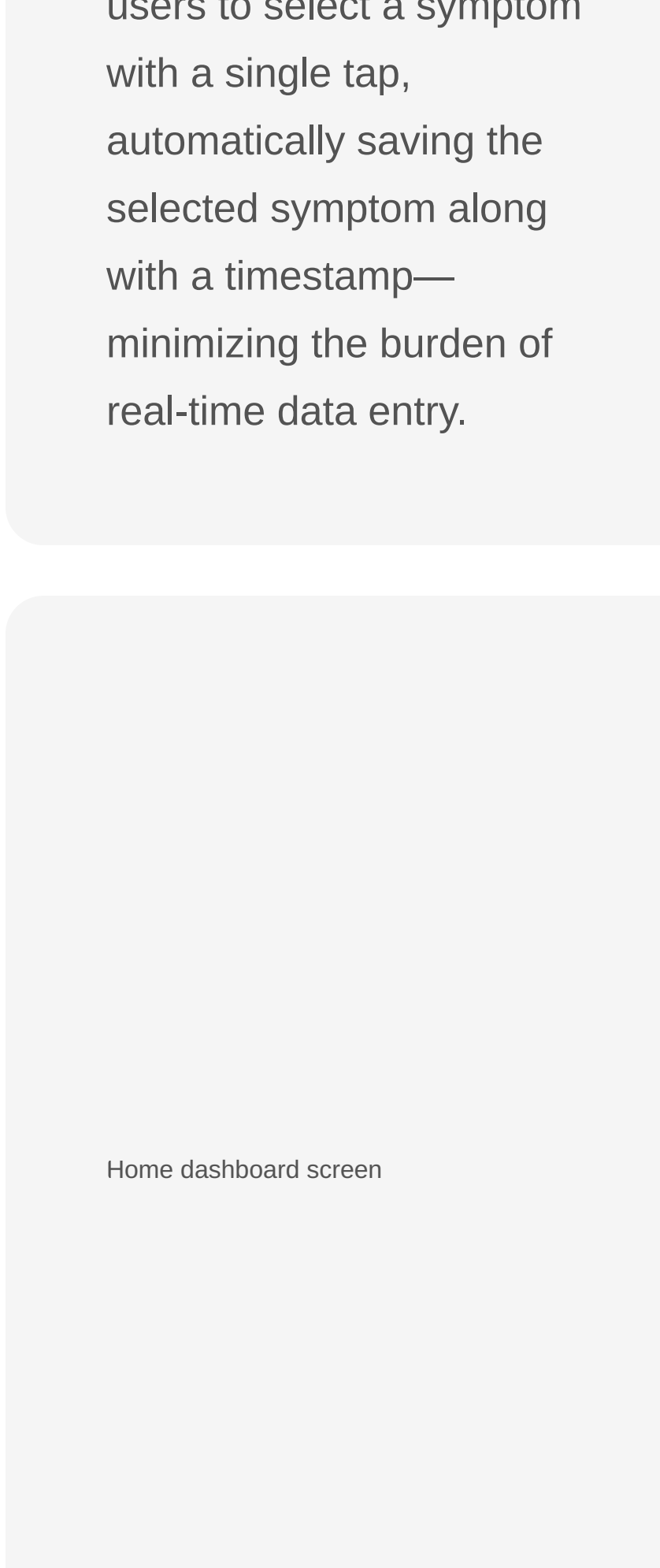
SOLUTION 1

Quick Logs During Daytime, Detailed Updates Afterwards

A screenshot of a mobile app interface for symptom input. It features a large, light blue button with the text "Log Symptom" in white. Below the button, there is a section titled "Symptoms" with a list of categories: "Anxiety", "Depression", "Mood swings", "Obsessive-compulsive", "Panic attacks", "Phobias", "Sleep problems", and "Tics". Each category has a corresponding icon and a brief description. The background is a soft, light blue gradient.

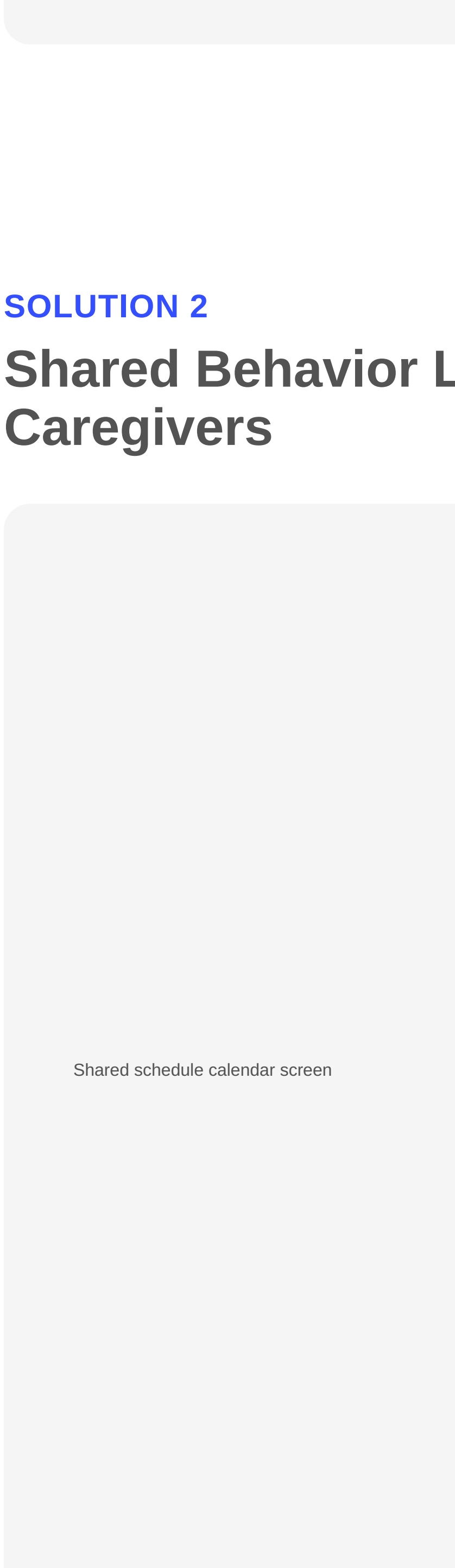
Symptom input during sign-up

Children with developmental disabilities often display a recurring set of symptoms. During sign-up, users can search for and select relevant symptoms that their child frequently exhibits. By categorizing these symptoms in advance, the system can later provide tailored content and personalized insights based on the selected symptom types.

A screenshot of a mobile app interface for quick logging. It shows a "Log Symptom" button at the top. Below it, there is a section titled "Quick Log" with a list of symptoms: "Anxiety", "Depression", "Mood swings", "Obsessive-compulsive", "Panic attacks", "Phobias", "Sleep problems", and "Tics". Each symptom has a corresponding icon and a brief description. The background is a soft, light blue gradient.

Quick logging during daytime

When a symptom occurs, caregivers are often focused on managing the situation and unable to document it in detail. The Quick Log feature allows users to select a symptom with a single tap, automatically saving the selected symptom along with a timestamp—minimizing the burden of real-time data entry.

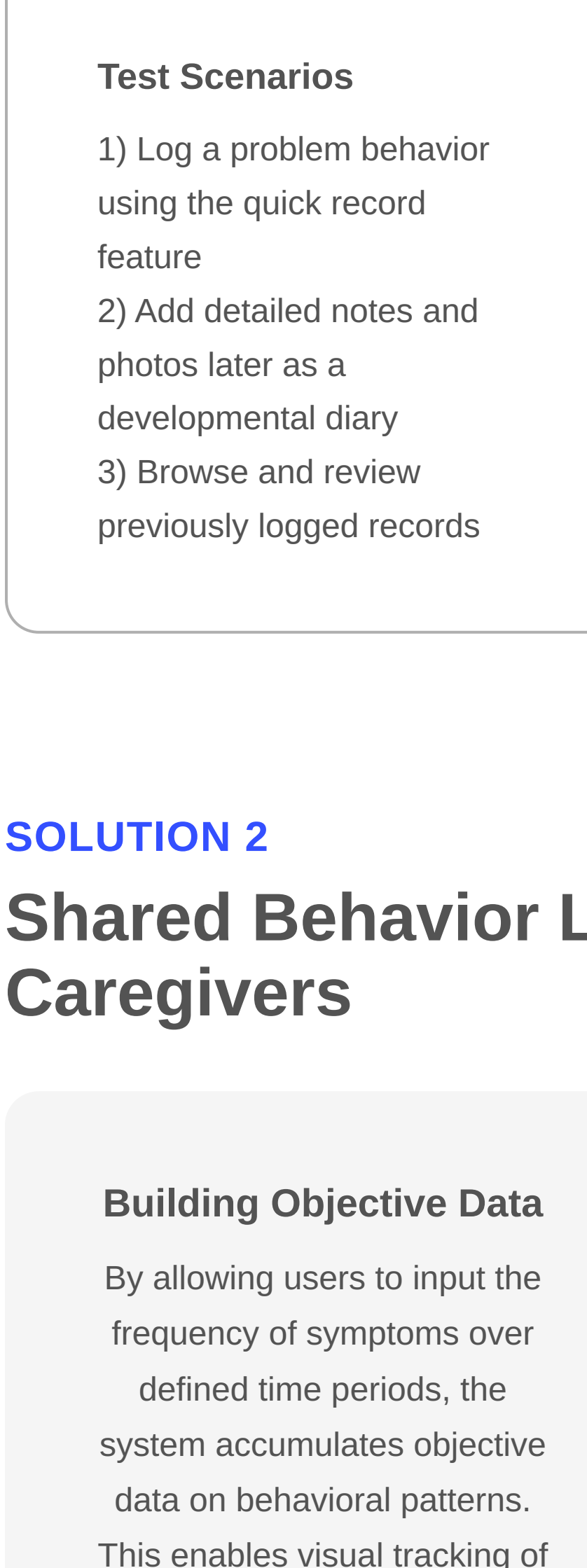
A screenshot of a mobile app interface for the home dashboard. It shows a "Log Symptom" button at the top. Below it, there is a section titled "Home Dashboard" with a list of symptoms: "Anxiety", "Depression", "Mood swings", "Obsessive-compulsive", "Panic attacks", "Phobias", "Sleep problems", and "Tics". Each symptom has a corresponding icon and a brief description. The background is a soft, light blue gradient.

Detailed context logging

Records created via the Quick Log can be expanded later to include contextual information such as triggers, consequences, and calming strategies. This helps analyze behavioral patterns in more depth and creates a valuable record for treatment and professional consultation.

SOLUTION 2

Shared Behavior Logging Across All Caregivers

A screenshot of a mobile app interface for a shared schedule calendar. It shows a calendar view with a list of dates. Each date has a corresponding icon and a brief description. The background is a soft, light blue gradient.

Logging Based on Shared Schedule

In most cases, parenting is not handled solely by the parents. Grandparents and childcare providers are often involved in the child's daily care. This leads to gaps in behavioral records and fragmented information. By linking behavior logging to a shared schedule, caregivers who spend time with the child are prompted to record observations in a timely and structured way.

My schedule

Others

Today

User Testing with Target Users

A core-feature MVP was developed as an Android app and tested with target users through user testing (UT) sessions.

User testing scenario screens

Test Scenarios

- 1) Log a problem behavior using the quick record feature
- 2) Add detailed notes and photos later as a developmental diary
- 3) Browse and review previously logged records

Step 2 (detailed input) was found to be quick and convenient. However, users found it difficult to locate and interpret past records intuitively.

SOLUTION 2

Shared Behavior Logging Across All Caregivers

Building Objective Data

By allowing users to input the frequency of symptoms over defined time periods, the system accumulates objective data on behavioral patterns. This enables visual tracking of symptom progression for caregivers, while also laying the foundation for personalized service expansion.

Focused View for Clinical Use

The reporting feature prioritizes delivering only clinically relevant information in a short time frame. Informed by physician interviews, it allows filtering by symptom and time to surface trends in frequency and severity.

A screenshot of a mobile app interface for response reports. It shows a line chart with a title "Response reports - behavioral patterns chart". The chart has a y-axis labeled "Frequency" and an x-axis labeled "Time". The chart shows a line that starts at a low frequency and increases over time. The background is a soft, light blue gradient.

A screenshot of a mobile app interface for response reports. It shows a list of records with a title "Response reports - all records list". The list has columns for "Symptom", "Frequency", and "Time". The background is a soft, light blue gradient.